Suggested problems: 1-5, 6(a), 7, 8

Problem 1: (Example 3.4.14) Suppose that X and Y are jointly continuous random variables with supports in the interval $[0, 2]$. Suppose also that the joint cumulative distribution function has

$$F(x, y) = \frac{1}{16}xy(x + y)$$

for $0 \leq x \leq 2$ and $0 \leq y \leq 2$.

(a) Compute a formula for the cumulative distribution function for every point (x, y) in the plane.

(b) Compute the cumulative distribution functions of X and Y.

(c) Compute the joint probability density function of X and Y.